# Customer Shopping Behavior Analysis

## 1. Project Overview

This project analyzes retail customer purchasing behavior to identify trends that improve sales, customer engagement, and loyalty. It focuses on understanding how factors like discounts, reviews, seasons, and sales channels (online/offline) influence buying decisions.

## 2. Dataset Summary

-Rows: 3,900

- Columns: 18

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

-Missing Data: 37 values in Review Rating column

## 3. Exploratory Data Analysis (EDA) – Python

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 3900.000000 | 3900.000000 | 3900 | 3900 | 3900 | 3900.000000 | 3900 | 3900 | 3900 | 3900 | 3863.000000 | 3900 | 3900 | 3900 | 3900 |
| unique | NaN | NaN | 2 | 25 | 4 | NaN | 50 | 4 | 25 | 4 | NaN | 2 | 6 | 2 | 2 |
| top | NaN | NaN | Male | Blouse | Clothing | NaN | Montana | M | Olive | Spring | NaN | No | Free Shipping | No | No |
| freq | NaN | NaN | 2652 | 171 | 1737 | NaN | 96 | 1755 | 177 | 999 | NaN | 2847 | 675 | 2223 | 2223 |
| mean | 1950.500000 | 44.068462 | NaN | NaN | NaN | 59.764359 | NaN | NaN | NaN | NaN | 3.750065 | NaN | NaN | NaN | NaN |
| std | 1125.977353 | 15.207589 | NaN | NaN | NaN | 23.685392 | NaN | NaN | NaN | NaN | 0.716983 | NaN | NaN | NaN | NaN |
| min | 1.000000 | 18.000000 | NaN | NaN | NaN | 20.000000 | NaN | NaN | NaN | NaN | 2.500000 | NaN | NaN | NaN | NaN |
| 25% | 975.750000 | 31.000000 | NaN | NaN | NaN | 39.000000 | NaN | NaN | NaN | NaN | 3.100000 | NaN | NaN | NaN | NaN |
| 50% | 1950.500000 | 44.000000 | NaN | NaN | NaN | 60.000000 | NaN | NaN | NaN | NaN | 3.800000 | NaN | NaN | NaN | NaN |
| 75% | 2925.250000 | 57.000000 | NaN | NaN | NaN | 81.000000 | NaN | NaN | NaN | NaN | 4.400000 | NaN | NaN | NaN | NaN |
| max | 3900.000000 | 70.000000 | NaN | NaN | NaN | 100.000000 | NaN | NaN | NaN | NaN | 5.000000 | NaN | NaN | NaN | NaN |

| Previous Purchases | Payment Method | Frequency of Purchases |
|---|---|---|
| 3900.000000 | 3900 | 3900 |
| NaN | 6 | 7 |
| NaN | PayPal | Every 3 Months |
| NaN | 677 | 584 |
| 25.351538 | NaN | NaN |
| 14.447125 | NaN | NaN |
| 1.000000 | NaN | NaN |
| 13.000000 | NaN | NaN |
| 25.000000 | NaN | NaN |
| 38.000000 | NaN | NaN |
| 50.000000 | NaN | NaN |

- Missing Value Treatment
  - Identified missing values using df.isnull().sum().
  - Imputed missing review_rating values using category-wise median.
- Column Standardization
  - Renamed columns to snake_case format for consistency.
- Feature Engineering
  - Created age_group for customer segmentation.
  - Created purchase_frequency_days to measure buying behavior.
- Data Consistency
  - Removed redundant column promo_code_used.

## 4. Database Integration

- Exported cleaned data to SQL database.
- Created structured tables (customers, transactions, products, etc.) with proper keys.
- Performed queries to analyze revenue, customer retention, repeat purchases, and discount impact.

This ensured a clean, structured, and business-ready dataset for dashboard development.

## 5. Data Analysis using SQL (Business Transactions)

Using **MySQL Workbench**, we executed structured SQL queries to analyze customer behavior, revenue trends, and key performance indicators.

**1. Revenue by Gender** – Compared the total revenue generated by male and female customers.

| | gender | revenue |
|---|---|---|
| 1 | Female | 75,191 |
| 2 | Male | 157,890 |

**2.High-Value Customers Utilizing Discounts** – Customers who applied promotional discounts yet recorded purchase amounts above the overall average.

| | customer_id | purchase_amount_ |
|---|---|---|
| 1 | 2 | 64 |
| 2 | 3 | 73 |
| 3 | 4 | 90 |
| 4 | 7 | 85 |
| 5 | 9 | 97 |
| 6 | 12 | 68 |
| 7 | 13 | 72 |
| 8 | 16 | 81 |
| 9 | 20 | 90 |
| 10 | 22 | 62 |
| 11 | 24 | 88 |
| 12 | 29 | 94 |

**3. Top 5 Products by Rating –** Identified the five highest-rated products based on average customer review scores.

| | item_purchased | Average Product Rating |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.8 |
| 5 | Skirt | 3.78 |

**4.Shipping Type Comparison –** Compared average spending across different shipping methods to understand customer purchase behavior.

| | shipping_type | avg_purchase_amount |
|---|---|---|
| 1 | Express | 60.48 |
| 2 | Free Shipping | 60.41 |
| 3 | Next Day Air | 58.63 |
| 4 | Standard | 58.46 |
| 5 | 2-Day Shipping | 60.73 |
| 6 | Store Pickup | 59.89 |

**5. Subscribers vs. Non-Subscribers –** Compared average spend and total revenue across subscription status.

| | Subscription_Status | Total Customers | Average Spend | Total Revenue |
|---|---|---|---|---|
| 1 | Yes | 1053 | 59.49 | 62645 |
| 2 | No | 2847 | 59.87 | 170436 |

**6. Discount-Dependent Products –** Identified 5 products with the highest percentage of discounted purchases.

| | item_purchased | Discount Percentage |
|---|---|---|
| 1 | Hat | 50.00 |
| 2 | Sneakers | 49.66 |
| 3 | Coat | 49.07 |
| 4 | Sweater | 48.17 |
| 5 | Pants | 47.37 |

**7. Customer Segmentation –** Classified customers into New, Returning, and Loyal segments based on purchase history

| | customer_segment | Number of Customers |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | Returning | 701 |
| 3 | New | 83 |

**8. Top 3 Products per Category –** Listed the most purchased products within each category.

| | Category Rank | Category | Item_Purchased | Total Sales |
|---|---|---|---|---|
| 1 | 1 | Accessories | Jewelry | 171 |
| 2 | 2 | Accessories | Sunglasses | 161 |
| 3 | 3 | Accessories | Belt | 161 |
| 4 | 1 | Clothing | Blouse | 171 |
| 5 | 2 | Clothing | Pants | 171 |
| 6 | 3 | Clothing | Shirt | 169 |
| 7 | 1 | Footwear | Sandals | 160 |
| 8 | 2 | Footwear | Shoes | 150 |
| 9 | 3 | Footwear | Sneakers | 145 |
| 10 | 1 | Outerwear | Jacket | 163 |
| 11 | 2 | Outerwear | Coat | 161 |

**9. Repeat Buyers & Subscriptions –** Checked whether customers with more than five purchases are more likely to subscribe.

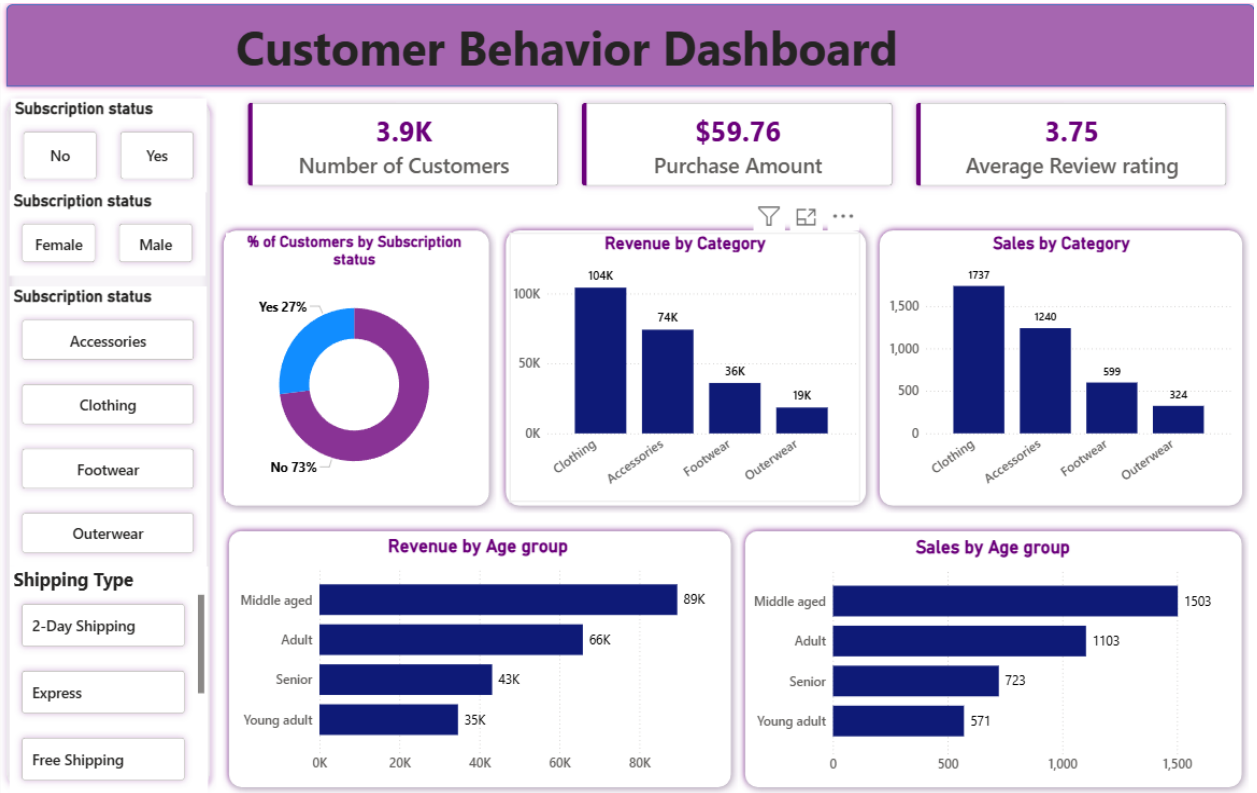
| | Subscription_Status | Number of Repeat Buyers | Percentage |
|---|---|---|---|
| 1 | Yes | 958 | 27.56 |
| 2 | No | 2518 | 72.44 |

**10. Revenue by Age Group –** Calculated the total revenue contribution of each age group to analyse spending patterns across different age segments.

| | age_group | total_revenue |
|---|---|---|
| 1 | Middle aged | 89,445 |
| 2 | Adult | 65,842 |
| 3 | Senior | 43,164 |
| 4 | Young adult | 34,630 |

## 6. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.




## 6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the "Loyal" segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.